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## Unit 3 : Session 8 : SLO 1 | Apply Exception Handling in Database Access

### I. Fill in the Blanks

1. SQLException is the base checked exception for all JDBC database access errors.
2. finally the finally block (or try-with-resources) ensures database resources are closed properly.
3. AutoCloseable try-with-resources automatically closes any object that implements the AutoCloseable interface.
4. subclass superclass (inheritance) in a multi-catch block the exception types must not be related by inheritance.

### II. True or False

1. TRUE SQLException provides getMessage(), getSQLState() and getErrorCode() for diagnosing database errors.
2. TRUE ResultSet and Statement must be explicitly closed in a finally block, or automatically via try-with-resources.

### III. Multiple Choice

1. B catch(SQLException e) is the correct syntax for catching database-related exceptions.
2. C getSQLState() returns a standardized five-character SQL error code string.

### IV. Programming

1. Identify the output: A The connection fails, the SQLException is caught, and the program prints "Error: " followed by the driver-supplied message (e.g. invalid URL / connection refused).
2. Identify the error: B executeQuery() throws SQLException, not IOException; the catch clause must declare SQLException (or Exception), otherwise the program does not compile.